LAB | Data Lake, Data Warehouse, and Lakehouse Architecture

Week 2 — Day 4: Data Lake, Data Warehouse, and Lakehouse Architecture — Lab

BritMart is deciding where different kinds of data should live and how that data should move from raw input to trusted business output.

This lab focuses on architecture decisions, schema timing, Bronze/Silver/Gold layers, compute and storage, and basic metadata. It uses only concepts covered today and skills already covered earlier.

Part 1 — Data Lake, Data Warehouse, and Lakehouse

Q. 1 — Choose the Better Platform

For each requirement, choose **data lake** , **data warehouse** , or **lakehouse** .

Requirement	Choice	Reason
Keep large amounts of raw structured and semi-structured data	Data Lake	Data Lake provides flexibility for structured and unstructured data. Storing large amounts of data is convenient and cheap as compared to Data Warehouse.
Give analysts controlled SQL-ready business tables	Data Warehouse	Suitable for structured tabular repository, consolidates data in well-organized manner for business and analytical purposes.
Keep flexible storage while also supporting governed table-style analytics	Data Lakehouse	One platform that unifies all data and applies data governance, thus avoiding Data Swamping.
Keep raw website events for future use	Data Lake	Raw events can be stored in raw or unprocessed format.

Q. 2 — Place BritMart Data

BritMart has:



MySQL order extract
supplier CSV files

> website JSON events
> monthly finance table
> daily sales summary

Place each item where it would most naturally fit in a data lake, warehouse, or lakehouse design.

There may be more than one valid answer. Explain your choice.

Solution:

Data	Storage	Reason
MySQL order extract	Data Lake, Data Lakehouse	can store raw data originating from the source.
supplier CSV files	Data Lake, Data Lakehouse	can store raw data originating from the source.
website JSON events	Data Lake, Data Lakehouse	unstructured data can be stored in Silver layer
monthly finance table	Data Warehouse, Data Lakehouse	structured historical data
daily sales summary	Data Warehouse, Data Lakehouse	structured historical data

Q. 3 — Compare the Three Architectures

Complete the table.

Area	**Data Lake**	**Data Warehouse**	**Lakehouse**
Raw data support	Strong fit	Not suitable	Strong fit
Strong structured reporting	suitable but requires additional effort	suitable	suitable
Flexible data shapes	Strong fit	Not suitable, only structured	Strong fit
Table-style analytics	suitable but require other tools	suitable	suitable

Q. 4 — MCQ: Data Warehouse

Which requirement is the strongest fit for a data warehouse?

Solution: D. Run controlled business reporting over trusted structured data

Part 2 — Schema-on-Read, Schema-on-Write, Bronze, and Silver

Q. 5 — Schema-on-Read or Schema-on-Write?

Choose the best description.

💡 A. Data is stored first and interpreted when it is read.
B. Data must match an expected structure before it is accepted into the target.
C. A raw JSON area keeps different optional fields.
D. A controlled reporting table requires known columns and types.

Solution:

A. schema-on-read

B. schema-on-write

C. schema-on-read

D. schema-on-write

Q. 6 — Design Bronze and Silver Rules

BritMart receives:

💡 order_id
city
amount_gbp
status

Define what should happen in:

Bronze:

💡 What should be kept?
Solution: All the raw data needs to be kept as it is.

Silver:

💡 What should be checked or cleaned?
Solution: The raw data is then validated in this layer:

- Filters out missing values

- Removes duplicates
- Checks and lightly transforms data

Q. 7 — Process Three Batches from Bronze to Silver

Use Pandas.

Batch 1

💡 [
(101, "London", 45.50, "complete"),
(102, "Manchester", -5.00, "complete"),
]

Batch 2

💡 [
(103, " Bristol ", 30.00, "complete"),
(104, "Leeds", 20.00, "cancelled"),
]

Batch 3

💡 [
(101, "London", 45.50, "complete"),
(105, "Manchester", 51.00, "complete"),
]

Treat all input rows as Bronze.

Create Silver by:

- keeping positive amounts;
- trimming city names;
- removing duplicate `order_id` values.

Print Bronze and Silver row counts.

Solution: lab_assignment_day9.ipynb file

Q. 8 — MCQ: Silver

Which record belongs most naturally in Silver?

Solution: C. A cleaned and deduplicated order row

Part 3 — Gold, Data Types, and Storage-First Architecture

Gold contains business-ready output built from trusted data.

Q. 9 — Build a Gold Summary

Using the Silver DataFrame from Q. 7:

- keep only completed orders;
- calculate revenue by city;
- sort highest revenue first.

Solution: lab_assignment_day9.ipynb file

Q. 10 — Structured, Semi-Structured, or Unstructured?

Classify:

> 💡 MySQL orders table
> JSON website event
> CSV supplier file
> product photo
> PDF supplier contract

Solution:

MySQL orders table : structured
JSON website event: semi-structured
CSV supplier file: structured with tabular data/ semi-structured with complex structures
product photo: unstructured
PDF supplier contract: unstructured

Q. 11 — Design a Storage-First Flow

BritMart wants to keep received data before processing it.

Draw:

> 💡 Sources
> Storage
> Processing
> Trusted output

Show at least three sources.

Solution:

A['My SQL'] ——→ S['Storage']

B['CSV file'] →S['Storage']

C['JSON file'] → S['Storage']

S['Storage'] ——→ P['Processing']

P['Processing'] ——→ T['Trusted Output']

Q. 12 — MCQ: Gold

Which output is the strongest Gold example?

Solution: A. Daily revenue by city for reporting

Part 4 — Compute, Storage, Metadata, and Catalogs

Q. 13 — Compute or Storage?

Classify each item.

> 💡 A. Files kept for later use
> B. CPU and memory used to run a transformation
> C. Stored analytical table data
> D. Processing resources used for a query

Solution:

A. Storage

B. Compute

C. Storage

D. Compute

Q. 14 —Why Separate Compute and Storage?

BritMart stores data all day but runs a large transformation only at night.

Explain why being able to change compute without moving all stored data can be useful.

Solution:

The separation allows compute to scale independently from stored data.

Benefits may include:

- different compute for different workloads;
- shutting down compute when not needed;
- shared access to the same stored data.

Q. 15 — Create a Small Metadata Catalog

For these datasets:

> 💡 bronze_orders
> silver_orders
> gold_daily_sales

Create a table with:

> 💡 dataset name
> purpose
> main columns
> source
> owner/team
> last known update

The purpose of metadata here is to help people understand what a dataset contains and how it should be used.

Solution:

dataset	purpose	main columns	source	owner/team	last known update
bronze_orders	raw order data	order_id, city, amount, status	batch	data engineer	2026-10-15

silver_orders	transformed clean data	order_id, city, amount, status	bronze_orders	data engineer	2026-10-15
gold_daily_sales	analytical ready data	city, revenue	silver_orders	data analyst, data scientist	2026-10-15

Q. 16 — MCQ: Metadata

Why is metadata useful?

Solution: B. It describes data so people and systems can understand what the data represents.